

Problem Set 3

Tim Price · 2026-10-02

1. It takes approx. 15.728 ms to load every page entry, which is approx. 13.1% of the process's quantum. For the quantum to spend approx. ten percent of its time loading the page table, the quantum would need to be approx. 157.28 ms long.

$16 \text{ KiB} = 2^{14}$
 $32 - 14 = 18 \text{ byte offset}$
 $2^{18} = 262,144 \text{ page table entries}$
 $(262,144) 60 \text{ ns} = 15,728,640 \text{ ns} = 15.728 \text{ ms}$
 b/c 32-bit address space loads every 60 ns $\rightarrow$ to load every page entry
 $0.13106 \approx 13.1\%$
 $15.728 / 120 = 13.106\%$ of the page's quantum is spent loading the page table
 $15.728 = 120 \cdot (0.13106)$
 $\frac{15.728}{0.13106} = 120 \text{ ms} \Rightarrow$ ratio of page load and percentage of time quantum spent loading.
 $\frac{15.728}{0.1} = 157.28 \text{ ms}$ quantum would make page table load approximately 10% of available time.

Figure 1: solution proof - problem 1

2.

- A.
- B.